Garden Guard

STM32L4S5 game | Native laptop display + sensing + player interaction

S1 AY2026/27 - ASSIGNMENT 2

Protect your patch.
Save the humans.
Rescue the drone.

Game enhancement built on every compulsory behavior.

6	zombie hits to win
3	starting lives
4 s	per displayed target

Group [NUMBER] | [Member 1 Name / Matric] | [Member 2 Name / Matric]

Recorded laptop display: about 29.5 fps. D6/PB1 buzzer confirmed audible on 24 Oct 2026.

Original C renderer runs on the MCU. Physical rescue, timeout and a full winning round remain pending.

The compulsory baseline stays intact.

The animated game adds presentation, scoring and feedback to the specified drone behavior.

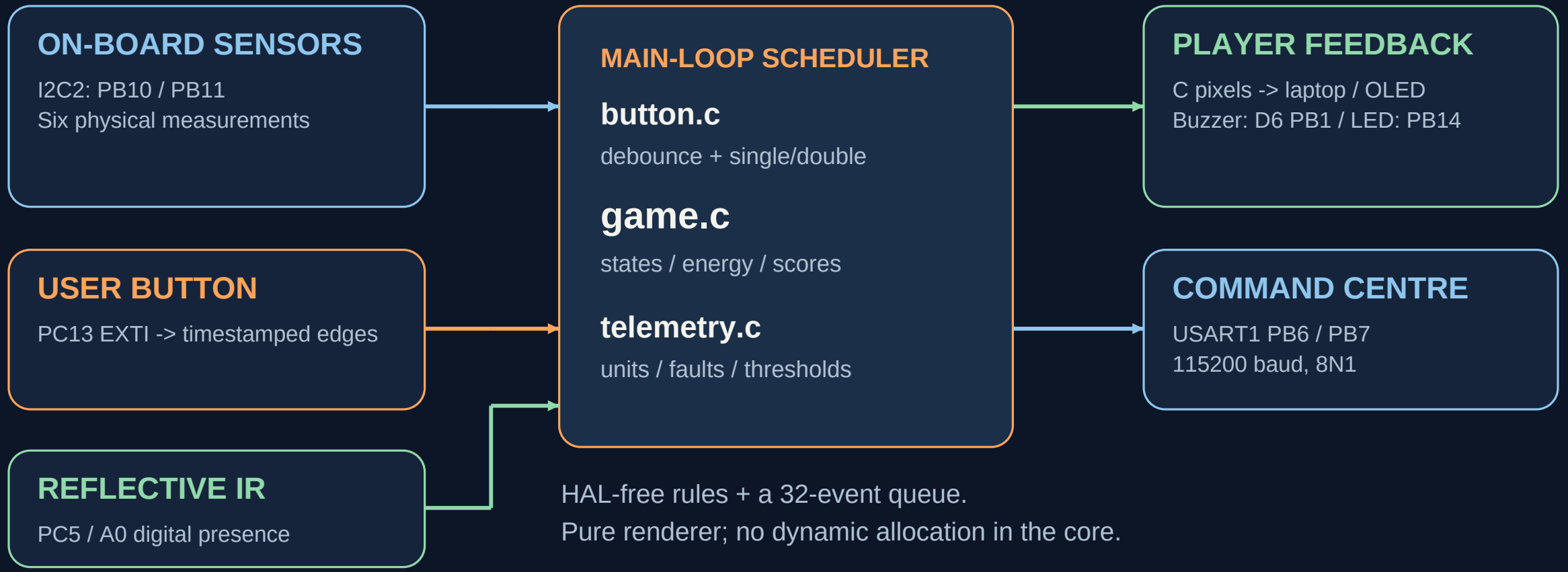
Six sensors	Read all six each second; report physical quantities with units.	IMPLEMENTED
UART monitoring	Normal telemetry OR all threshold/fault logs; automatically resume.	IMPLEMENTED
Interrupt pushbutton	Single: +3/10 energy. Double (<500 ms): mode switch or rescue.	IMPLEMENTED
LED and automatic fire	Standby ON; battle 1 Hz; attacked 2 Hz. Fire at energy >=5/10.	IMPLEMENTED
Attacked / rescue	10 s rescue window; expiry stops acquisition and application output.	IMPLEMENTED

Physical checks are partial; attack/rescue, calibration and a full winning demo remain pending.

Source: supplied assignment PDF, pp. 2-5; Core/Src/game.c and telemetry.c

A portable game core, bounded board I/O.

Separate the state machine from HAL peripherals so rules can be tested on the Mac.

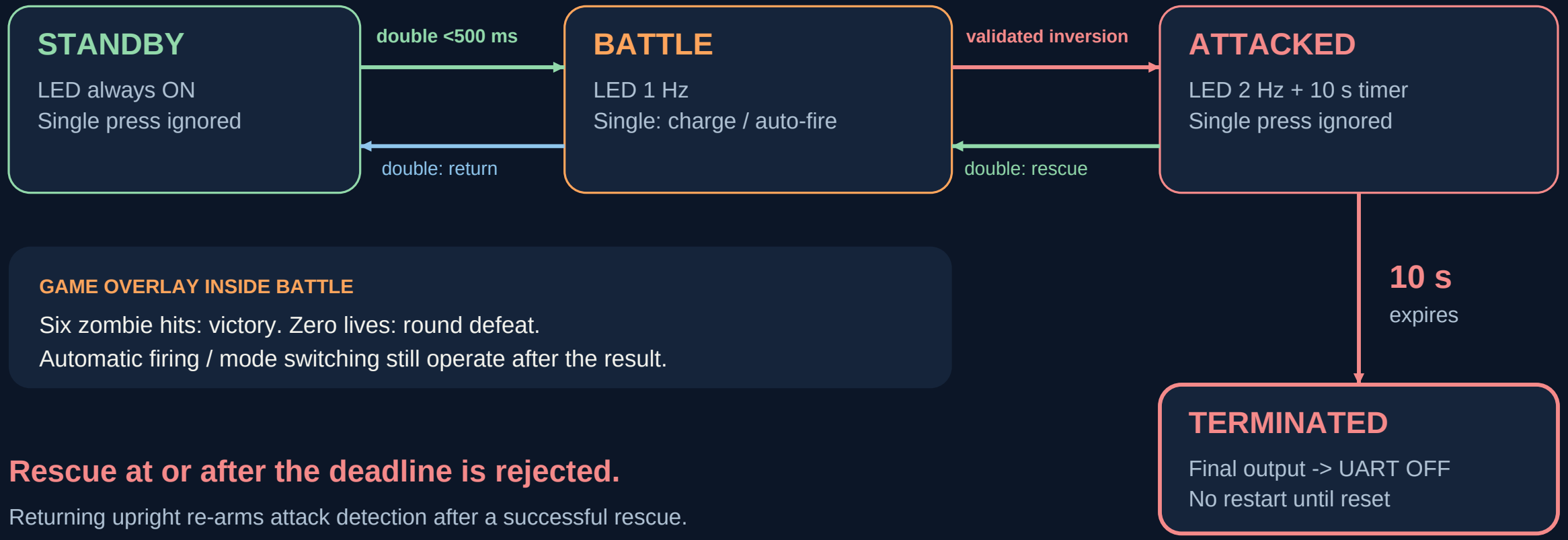


HAL-free rules + a 32-event queue.
Pure renderer; no dynamic allocation in the core.

Source: Core/Src/main.c, game.c, button.c, board_io.c, external_io.c, game_display.c

The rescue deadline is a terminal boundary.

Round victory/defeat remains a game overlay; it does not replace the compulsory mode controls.



GAME OVERLAY INSIDE BATTLE

Six zombie hits: victory. Zero lives: round defeat.
Automatic firing / mode switching still operate after the result.

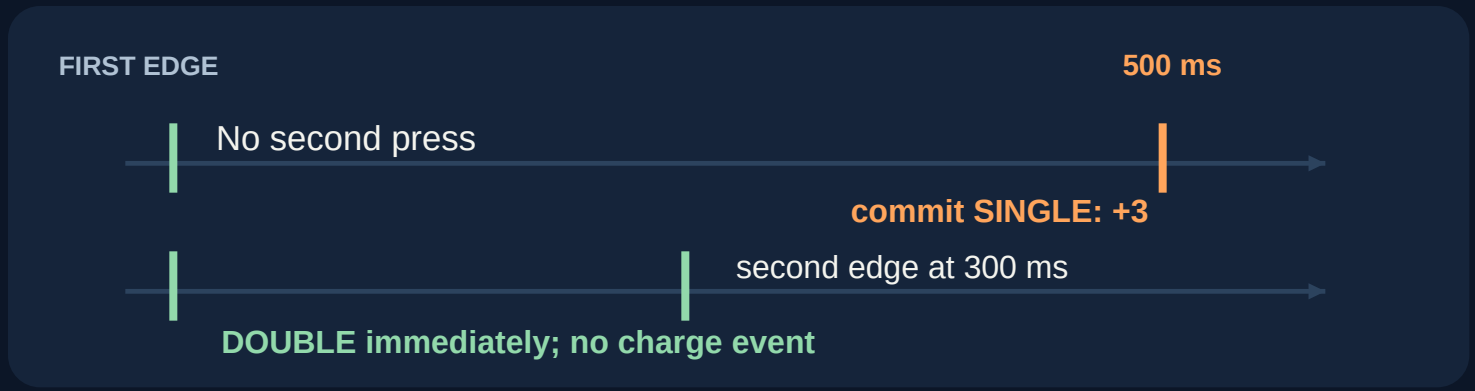
Rescue at or after the deadline is rejected.

Returning upright re-arms attack detection after a successful rescue.

Source: assignment PDF pp. 2-5; game_button(), game_tick(), game_update_sensors()

One button, unambiguous timing.

Delay a single action until its double-press window has closed; never charge on the first half of a double.



COMMITTED SINGLE PRESSES

0 -> 3 -> 6 -> 1

Second single reaches 6/10, auto-fires, then consumes 5/10.

ZOMBIE HIT

+10 points

+5 with IR aim lock

HUMAN AVOIDED

+5 points

Wait out its 4 s window

HUMAN HIT

-5 points / -1 life

Do not charge into a shot

ZOMBIE MISSED

-5 points / -1 life

No shot within 4 s

Six zombie hits wins; start with three lives. Exactly 500 ms is not a double press.

Source: button.c; game_config.h; game_button(); 30 ms debounce, strictly <500 ms double press

Six sensors, physical units, explainable limits.

Starting thresholds are configurable and must be calibrated under the actual demo conditions.

SENSOR	OUTPUT	NORMAL RANGE	INTERPRETATION
Temperature	degC	10 to 45	Operational environment
Pressure	hPa	850 to 1100	Atmospheric pressure
Humidity	%RH	20 to 85	Relative humidity
Accelerometer	m/s^2, magnitude	0 to 30	Motion / impact warning
Gyroscope	deg/s, magnitude	0 to 250	Rapid rotation warning
Magnetometer	uT, magnitude	0 to 200	Magnetic disturbance

UART at 1 Hz

Normal: T / P / H / A / G / M with units.

Any violation or invalid sensor: send all warning/fault lines instead; resume when clear.

Bench: all six valid; magnetic warning -> full normal UART recovered. Sensor accuracy is uncalibrated.

Source: app_config.h; board_io.c calibrated conversions; Telemetry_Check() / Telemetry_Format()

Use gravity for orientation; gyro for rotation.

A gyroscope alone reports angular rate, not a stable absolute upside-down direction.

1 / CAPTURE UPRIGHT

Hold still at reset.
6-13 m/s^2 gravity;
gyro <=20 deg/s.

2 / CONFIRM A FLIP

Integrate angular speed.
Require >=60 degrees
of measured rotation.

3 / HOLD INVERSION

Gravity opposes baseline
(cosine <=-0.7)
for at least 350 ms.

4 / ENTER ATTACKED

Start the 10 s timer.
Double press rescues;
upright re-arms detection.

REJECT BAD EVIDENCE

Invalid motion samples or gaps >250 ms reset continuity. Strong non-gravity acceleration is rejected.
Hysteresis and an upright re-arm prevent repeated attacks while the board stays inverted.

Source: game_update_sensors(); game_config.h; sampled every 50 ms

The MCU renders; the laptop shows its pixels.

Original 1-bit artwork in a native window. Physical button, sensors and buzzer remain on the board.

WALKING ZOMBIE

FLYING PEA / HIT FEEDBACK

FRIENDLY HUMAN - HOLD FIRE

PURE SOFTWARE RENDERER

game_display.c draws the 128x64 framebuffer.
Score, heart lives and energy bar stay visible.
Pea flight, recoil, impact and result effects.
Host previews show implementation, not a board pass.

LIVE DISPLAY AND PHYSICAL I/O

33 ms MCU frames; laptop targets 30 fps.
Aligned SWD reads; coherent sequence checks.
IR A0/PC5; buzzer D6/PB1 via TIM2 GPIO.
OLED I2C online; image remains distorted.

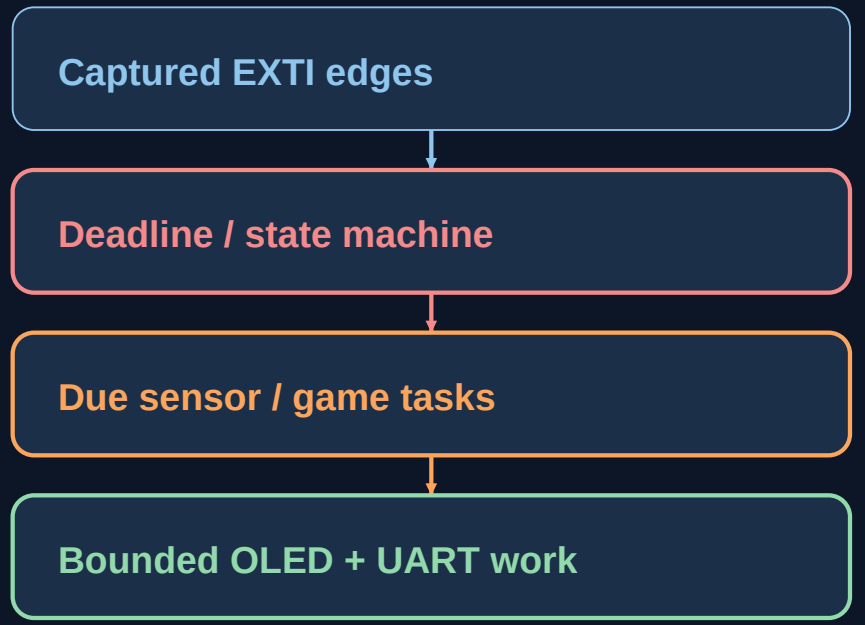
Monochrome shapes and FIRE / HOLD FIRE cues identify targets; the reflective module cannot classify colour.

Sources: game_display.c; preview_display.py host-rendered frames; app_config.h; external-hardware.md

SysTick drives deadlines and bounded work.

EXTI records edges. UART uses short HAL_UART_Transmit() calls; the scheduler does not use HAL_Delay().

TASK	CADENCE / BOUND
Motion + orientation	50 ms
All six sensors / UART	1000 ms
C framebuffer render	33 ms requested
IR stability filter	30 ms
External I2C timeout	8 ms OLED / 5 ms

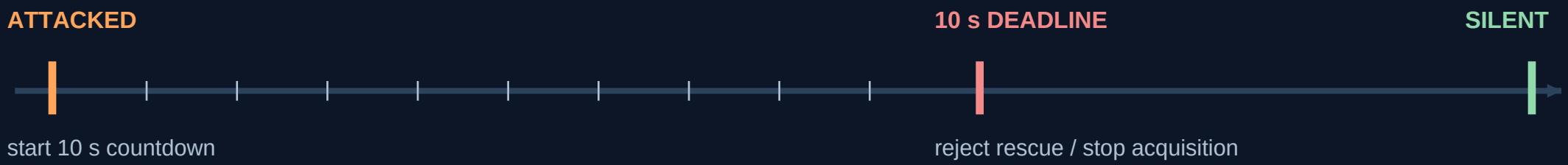


Laptop: 30 fps target. OLED: immutable 16-byte chunks at 100 kHz; recovery after 1 s if previously online.
UART: 16-byte foreground chunks, 2 ms HAL timeout; IRQs stay enabled. No HAL_Delay() scheduling.

Source: main.c due(); SysTick_Handler(); board_io.c UART queue; external_io.c OLED / note sequencer

A failed rescue ends the session predictably.

The game deadline wins over late input; shutdown finishes only queued output before communication stops.



Attacked log every second; double press before expiry rescues.

AT LOGICAL EXPIRY

Discard obsolete game chatter.

Queue final defeat / termination output.

No further sensor acquisition or game actions.

FINAL DRAIN: 600 ms BUDGET

Only UART + final OLED frame advance.

Then stop UART / buzzer / LED / interrupts.

CPU waits until a hardware reset.

Final frame stops laptop polling; UART drain then deinitialization establishes application silence.

Source: game_tick() expiry; shutdown_system(); Board_UartIdle(); External_FinishDisplay()

Software checks and selected bench observations.

D6 sound and native laptop display work; latest OLED communication passed but the image is distorted.

239

game / button checks

8,346

display renderer checks

29.5 fps

recorded laptop display

30 fps target; 29.4 fps received in this capture

VERIFIED IN SOFTWARE

Timing / rescue / energy / scores / rollover.
Renderer checks; visual events / ordering / wrap.
Telemetry units, faults and recovery: PASS.
Foreground scheduler scenarios: PASS.
Address / undefined-behavior sanitizers.

OBSERVED ON THE BOARD

24 Oct: audible D6 buzzer; live laptop frames.
OLED: 8 s capture, no faults; distorted image.
Injected button edges: zombie / human hits.
Earlier: physical double start / LED / charging.
Earlier: six sensors; IR present and absent.

Pending: full manual win, IR-assisted hit, physical rescue/timeout, measured LED/button timing and calibration.

Sources: host tests; 24 Oct UART / OLED records; output/laptop-display/verification-30fps.json

Bench findings guide the next integration step.

Corrected connectors restore sound and I2C; the laptop makes the original artwork readable.

Buzzer connector corrected	Latest D6 is PB1. TIM2 IRQ toggles this GPIO; user confirmed audible feedback on 24 Oct.
Reflective IR response verified	Held card: PC5 LOW / aim true. Removed card: PC5 HIGH / aim false. Active-low digital input confirmed.
OLED moved from D3 to I2C	Address 0x3c responded; eight seconds had no transfer faults. Pixels remain distorted; panel damage unproven.
Slow laptop framebuffer reads	Aligned 32-bit reads cut measured capture time from 92.75 ms to 8.36 ms; the window reached about 29.5 fps.
Magnetic warning and recovery	Above 200 uT: warnings. Below 200 uT: full telemetry resumed at 10:17:58. Baseline remains uncalibrated.

Communication success does not prove OLED image quality; the laptop uses the MCU's actual framebuffer.

Source: hardware-test-2026-10-22.md; animation-test-2026-10-22.md; external-hardware.md; I/O review

Proposed 20-minute presentation and demo.

Rehearse the full 20-minute assignment plan and a compressed 15-minute Lab 5 variant; retain Q&A readiness.

0-4 min	**Design**	Idea, baseline, architecture and state flow
4-8 min	**Controls + game**	Laptop pixels, +3 charge, auto-fire, human avoidance
8-12 min	**Sensors + UART**	Six units, thresholds, fault log and recovery
12-16 min	**Attack + rescue**	Physical flip, 2 Hz LED, rescue double press
16-20 min	**Failure + discussion**	10 s timeout, UART silence/reset, evidence and limits

Assignment brief governs the full plan. Confirm the assessed slot; record actual outcomes.

Sources: Assignment 2 PDF pp. 5-6 (20 min); Lab 5 p. 18 (15 min reference variation)

Complete the remaining hardware demonstration.

Use the native laptop display for readable game pixels; assess OLED image quality separately.

- **Check display image and extended stability**

  I2C now responds. Inspect distorted OLED pixels separately; retain a longer UART/display recording.

- **Exercise animation and aim bonus**

  Check pea/hit effects and HUD; confirm IR presence adds its bonus to a real zombie hit.

- **Play a complete manual winning round**

  Injected hits were captured; now use the real button to confirm sounds, scoring and the victory screen.

- **Validate every compulsory state**

  Measure LED/button timing; physical flip, rescue, timeout silence and reset recovery.

- **Calibrate, capture and finalize**

  Tune sensor/IR limits; retain UART/photos; fill group details and actual contribution appendix.

Keep code archive and slides separate; rename both with the required group / matriculation details.

References: ST UM2708; SSD1306 / sensor datasheets; external-hardware.md

Appendix: planned responsibilities, not a declaration.

Replace placeholders with actual individual and joint work before submitting or presenting.

[MEMBER 1 NAME / MATRIC] - PLANNED 50%

- Player interaction and game rules
- IR presence / optional colour module
- Button interrupts, charge / auto-fire
- Buzzer feedback and LED patterns

[MEMBER 2 NAME / MATRIC] - PLANNED 50%

- Display, sensors and monitoring
- OLED integration; six compulsory sensors
- Orientation, UART and thresholds
- Physical calibration and telemetry evidence

JOINT - PLANNED

Integration, rescue/termination, laptop viewer, hardware debugging, flowcharts, slides and demonstration.

Original assignment requires actual joint/individual contributions and allows PPT/PPTX/PDF/KEY. Lab 5 p. 18 calls the appendix optional and lists PPT only; follow the assignment brief.

AI-assisted draft. Planned percentages do not establish completed contributions; replace them with actual work.

ST board manual / UM2708 SSD1306 datasheet Optional TCS3472 datasheet

Source: assignment PDF pp. 5-6; proposed 50/50 split; hardware report. Actual contribution declarations pending.